

ISAAC ZOLANA

Software Developer

Sainte-Marthe-sur-le-Lac, QC | isaaczolana.dev@gmail.com

hizackdavid.github.io | github.com/HiZackDavid | linkedin.com/in/isaac-zolana

PROFESSIONAL SUMMARY

Software engineering student with internship experience in full-stack development, cloud infrastructure, DevOps, and AI/ML. Strong foundation in software development and algorithms, with hands-on experience building applications and cloud-based solutions using modern technologies.

TECHNICAL SKILLS

Programming Languages: Python, Java, C#, JavaScript, TypeScript

Frontend: React, Next.js, Angular, Blazor, Tailwind CSS

Backend: FastAPI, Spring Boot, ASP.NET, REST APIs

Cloud & DevOps: Azure, GCP, Docker, Kubernetes, Terraform, Azure DevOps, CI/CD

Databases: PostgreSQL, pgvector, MySQL, MongoDB, Redis

AI & Search: RAG, vector search, hybrid search, embeddings, Vertex AI

Mobile: React Native, Jetpack Compose, Flutter

Testing & QA: Playwright, Cypress, JUnit, Postman

Tools: Git, GitHub, Jira, Storybook, Figma, Firebase, Supabase, Unity

PROFESSIONAL EXPERIENCE

Software Developer Intern – AI/ML | Sep 2025 – Dec 2025

Airbus, Mirabel, QC

- Developed a RAG-based intelligent search engine proof of concept for Airbus technical publications.
- Built an XML preprocessing and indexing pipeline using dense and sparse embeddings with hybrid search.
- Used Vertex AI Workbench, Cloud Storage, Vertex AI Vector Search, PostgreSQL, and pgvector to support indexing and retrieval workflows.
- Presented technical progress during recurring demos and aligned priorities with stakeholders.

Software Developer Intern – DevOps & Infrastructure Intern | Jan 2025 – May 2025

Workleap, Montréal, QC

- Modernized deployment tooling and CI/CD pipelines across multiple services.
- Updated Azure DevOps pipelines for Kubernetes deployments across development, staging, and production environments.
- Migrated a project from Yarn to PNPM and updated scripts, documentation, and pipelines.
- Improved infrastructure security by moving application secrets to Azure Key Vault with Terraform-managed references.

Full-Stack Developer Intern | Aug 2023 – Apr 2024

Dcbel, Montréal, QC

- Developed web and mobile features using Blazor and .NET MAUI.
- Built reusable UI components and integrated frontend services with backend APIs.
- Implemented unit and end-to-end tests using Playwright.
- Designed custom Azure AD B2C authentication pages using HTML, CSS, and JavaScript.

QA & Software Development Intern | Jan 2022 – Apr 2022

Legalhood / OnRègle, Montréal, QC

- Performed manual QA across web services and documented usability, functionality, and workflow issues.
- Built Cypress end-to-end tests to improve regression coverage and product reliability.

Web Developer Intern | May 2019 – Dec 2021

Revenu Québec, Laval, QC

- Built dynamic Angular forms and reusable components for an internal service request application.
- Integrated frontend components with REST APIs while following UI/UX and accessibility standards.

EDUCATION

Bachelor of Software Engineering

École de technologie supérieure, Montréal, QC

Expected April 2027

Diploma of College Studies in Management Information Technology

Collège Montmorency, Laval, QC

2017 – 2020

SELECTED PROJECTS

SharedFlavors – Mobile Recipe Discovery App

Kotlin, Jetpack Compose, Firebase

- Developed an Android application for discovering, creating, and sharing recipes with authentication, profiles, maps, and administrative features.
- Built core application flows using Kotlin and Jetpack Compose while contributing through Git-based feature development and pull requests.

Huffman – Huffman Coding Implementation

Java, Algorithms, Data Structures

- Implemented Huffman coding in Java using tree-based data structures to encode and decode text.
- Applied priority queues, binary trees, and frequency-based encoding to implement lossless data compression.

ENGINEERING INVOLVEMENT

Dronolab – ÉTS Student Club

Member

- Contributed to a student engineering club focused on drone-related systems, technical collaboration, and multidisciplinary project work.

LANGUAGES

French – Fluent

English – Fluent

Portuguese – Conversational

Spanish – Conversational